

Meta Directory Services (MDS)

Achieving Identity Management through MDS

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Abstract

The objective of this research paper is to try to reveal what Meta Directory Services (MDS) are and the crucial role they have in an IT infrastructure and in Business. The paper mentions what qualities are required from a provider of MDS. The solutions provided by MDS ought to significantly reduce the costs of managing all the directories and databases in the enterprise and provide the enabling technology benefits of a Corporate Directory (CD) for B2B Commerce and eBusiness.

The paper also points out how Meta Directory Services (MDS) can provide a solution to many identity management needs. Thus, MDS used for dealing with the issues and problems surrounding disparate identity information is emphasized. A Meta Directory System enables the distribution of identity changes amongst various resources, ensuring that related data remains consistent throughout the enterprise and sharing of identity information among different resources.

The paper also will address the current prevailing issues of identity management and try to give an enterprise directory architecture model built upon MDS technologies that enables the central or remote administration of personnel information, identity, in dissimilar data repositories. The identity repositories will include Netscape's Directory Server, Microsoft Exchange, Lotus Notes, Oracle, People Soft, and NT's Primary Domain Controller (PDC) and Active Directory (DC).

Besides an identity management solution, MDS is able to create a corporate, enterprise wide, directory which is more of a corner stone of eBusiness infrastructure and B2B Commerce.

This paper lays out to the reader the advantage of using a MDS to improve business processes and IT to reduce costs and improve productivity.

Disclaimer

The Charleston Southern University Computing and Information Science department has supported and encouraged this research effort. The department with high caliber faculty encouraged and support all senior students to choose subject area of interest that will benefit them down the path in their career and make them well equipped when they get to the business world.

The primary focus will be on the identity management of personnel using a Meta Directory Service (MDS) as identity management solutions are discussed.

For further those interested in more in depth knowledge on the subject, information on books and web sites where this information can be found are made available.

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1.0 Introduction

Welcome to an analysis of Meta Directory Services (MDS). This research paper is made possible through a Computing and Information Sciences department of Charleston Southern University

The author attempts to address the need and business value of a MDS, what qualities are needed of a solution, what the offer in market is, and what the realistic goals ought to be in dealing with identity management. By Identity the author refers to the information about people, applications, and resources scattered in proprietary file formats, directories and databases throughout most information technology (IT) enterprises.

Because the subject of Identity management is a broad topic, it will not be covered entirely in this paper, though much of what will be covered directly addresses identity management issues and solutions.

There are many factors that have led to the growth of the MDS market in recent years:

- Business awareness of the growing complexity and cost of identity management in the enterprises today.
- Organizations realize the need for an Enterprise Application Integration (EAI) infrastructure to provide integration of both data and applications across multiple functions and increasingly multiple companies.
- Businesses are also looking to implement a Corporate Directory for B2B and eBusiness initiatives.

In large corporations steps have been taken to implement a Corporate Directory with the help of a MDS successfully. The availability of MDS based on technology standards, such as Lightweight Directory Access Protocol (LDAP) and Extensible Markup Language (XML), has made this long-term goal a realistic possibility.

Due to technology standards, such as LDAP and XML EAI are possible today. Experts have agreed that the best integration solutions are those that permit heterogeneous applications to participate in a shared, automated, end-to-end business process.

A capstone of eBusiness and B2B Commerce is Directory Services. With the rapid growth of the internet/intranet/extranet, the Directory Server market has also grown. Businesses need identity data to be readily available to a broad range of web-enabled application, which communicate over standard protocols such as LDAP, and use XML for data interchange. Directories server Network Operating Systems (NOS), Directory Enabled Networking (DEN), and a variety of middle ware products such as Single Sign-On applications and Application Servers. Most large software vendors are consolidating the identity repositories their applications use into directory services.

Vendor such as Netscape, Novell, Microsoft and IBM, to name a few, are all making efforts to build their own product lines around their own directory services. Netscape, now iPlanet/Innosoft, and Novell were one of the firsts to begin building its product around a directory server centric architecture.

A single directory would solve a wide range of the identity management problems most large organizations are facing today. Network Operating System (NOS) directories should be given special attention. This attention will determine a directory information tree (DIT) that is closely aligned with the actual physical network structure. And while the MDS can provision entries in the NOS directory, system administration is often performed directly within the NOS directory. Meta Directories represent all the employees in a corporation and not every employee in the company has a need for a network account.

A single directory, which all applications would access, sounds great but is unrealistic at this time for enterprises that include multiple vendors' products. At this time, the only realistic solution to the identity management problem is a MDS which can help facilitate the consolidation approach to a single directory but also maintain a synchronized multiple directory, multi-repository, environment, as well as enable a EAI, B2B and B2C infrastructure.

Why should organizations use a Meta Directory System solution?

Meta Directory System (MDS) solutions do not only simply synchronization data but implement business policies. They implement business rules and give a centralized administration of data to achieve enterprise focus.

Is it important to have data synchronization in an organization?

With data synchronized information is entered only once into any specified directory or application for it to be accessed across the entire network. For companies with multiple directories, synchronization reduces manual labor, minimizes opportunities for human error, and saves money that would be spent on redundant data management.

2.0 MDS Overview

Companies have struggled with the problem of identity management. The emergence of computing has brought in many different applications and identity repositories into the computing environments of today's companies. With the spread of networks, the data has also spread within them. As each of these applications were introduced, so was a new set of redundant data information that each application uses, each with multiple and differing data definitions

Identity is the summary of information about files, applications, resources scattered in directories, people, and databases throughout most IT enterprises

Today there are a various ways of dealing with identity management issues some of which will be discussed in this paper. Businesses organizations will need to evaluate their environments to assess the identity solutions that best meet their needs and also get the cost for putting it in place or not putting it in place as a solution. There will be long-run cost saving of even the simplest of solutions chosen.

Today businesses encounter a number of identity scenarios and are exploring ways for the most beneficial means of addressing these scenarios.

A Meta Directory System solution is the most cost effective means for identity solution and has many advantages.

The paper is going to address the business need and value of a Meta Directory solution, what qualities are needed in a solution, what is being offered in the market, and what the realistic goals should be in dealing with identity management. It also provides helpful information and guidelines for evaluating the enterprises' identity data.

2.1 Meta Directory Services (MDS)s as a means of Identity Management

MDS solutions reach far beyond just data synchronization. It creates a Meta Directory, which is an aggregate of all the identity repositories that are used to tie together the business infrastructure. The directories are helpful in implementing business policies and enforcement of business rules and also provide central administration to make sure an enterprise has is on track.

The value of MDS is shown or measured by the number of relationships the MDS manages between applications with a dissimilar data store and the number of applications that can leverage the corporate directory that MDS can create.

A sound and central means of identity management is requirement by the Enterprise user administration (EAU) to operate with great success.

As mentioned, MDS are used to produce a corporate directory service, a repository for eBusiness.

Typical e-business applications require that all the user entry data be stored on a single server in a single partition.

A MDS solution will provide the direction for a corporate IT infrastructure by provide the business process and principals needed to guide corporate IT decision-making. MDS will provide standard procedures for managing identity thus eliminating stovepipe solutions that are hard to maintain and expand.

To cost justify a MDS, it is necessary to start with an identity and application inventory. It may be found that a less costly management solution is adequate, alternative solutions will be provide, but it may be determined that a meta directory solution is most likely to meet the strategic business objectives.

2.2 Identifying Identity Management Needs

Enterprise Evaluation

In order to evaluate successfully, the following questions ought to be asked:

Is your network just a collection of different applications and systems? Do multiple directories and data systems exist in the organization? Are data objects and other users stored in multiple places in the system? How and what is the level of the environmental management? When there are new accounts how are they created for users each time a new application is added?

When a change is made does the event cause changes to occur in multiple locations and does it flow through the system to change? Is this change accomplished manually, or is it automated?

It is more than likely that the company is like most other companies. a lot of money is used in managing information that exists in multiple locations in the network. There are users with an identity in multiple applications and whenever a change occurs to that users' information, the administrator or authorized person has to duplicate the task of updating the information in many different places.

This problem is not uncommon to most companies today where there are multiple databases (directories) with redundant information.. it is also true that many businesses have not yet found out how to address the problem other than injecting more money to try to solve the ever increasingly complex problem.

Only 8% of large businesses have these fragmented data "islands" connected according to Forrester Research,. A great percentage of 42% try to synchronize the data manually! The rest are still trying to achieve a management solution, but still have a fragmented way of achieving it.

According to he Burton Group, a highly respected IT analyst and consulting firm provides an estimate of the management costs for a network with only 7 directories and 25,000 users. Based on standard calculations the redundant management costs account for \$312,600 of the total management cost. If the management of this information could be automated, then the cost would reduce from \$364,700 to \$52,100. Remember that other research indicates that most enterprise environments have many more directories...up to 181 different directories.

2.3 Identity Repository Assessment / Inventory

Taking stock of the inventory of the number of repositories and the resources needed to ensure that they are maintained worth doing. The results have encouraged enterprises to look at an Identity management solution. Based on this the cost can be justified. This process will reveal the cost of not implementing a Meta Directory Solution. The documentation of every cost of maintaining platforms and staff to manage all the disparate data repositories should be done.

2.4 What are the Initiatives an Identity Management Solution requires?

Due to the expanding nature of many businesses they are already starting to struggle with some form of identity management initiative to provide the enabling services needed by:

Global Address Book applications. Where synchronizing mailbox information between the different e-mail directories within a company enables users to locate other users and send them e-mail across different systems used by the company.

Hire/fire solutions. Disseminating information about a newly hired employee – like title, role and access rights to all systems that require identity data enables speedy establishment

of services, also known as **provisioning**. Systems must perform the same processes quickly in reverse when employees are fired to prevent breaches of security.

E-commerce applications. Synchronizing enterprise identity information is vital, such as digital certificates for suppliers and extranet users is enabled with directories that reside outside of firewalls.

Single Sign-On initiatives. Managing user password, access right, name, and information is enabled across many different platforms and applications that use a variety of access control and cryptographic techniques

3.0 Enterprise User Administration (EUA)

The Gartner research paper on "Enterprise User Administration (EUA) Magic Quadrant FY01" describes EUA in many of the same terms used here to describe MDS (MDS). Gartner states the EUA market is adapting beyond the traditional operating system and database environment and emerging as the key tool for user access management.

EUA — Also known as consolidated user administration or consolidated security administration — permits a security administrator (SA) to manage a number of permissions on behalf of a user. Adding a new user might be attained with one issue command. In practice, EUA tools cut down on the workload of the SA, but doesn't eliminate it. They also make the SA's job easier, and reduce turnaround time between request and implementation to less than 24 hours in some cases

Overview of EUA Tools

Customers will want an EUA product that has the following characteristic among many:

- Platform support — Operating system (OS), database management systems (DBMS), e-mail, groupware, Enterprise Resource Planning (ERP) and other purchased applications.
- Easy customization/integration with applications not supported "out of the box"
- Web application support, primarily through the management of extranet access management (EAM) products and LDAP
- LDAP management — Initially as a target platform, but increasingly as the repository for user credentials/entitlements
- Technical support — Breadth, depth and geographic availability
- Role-based access control covering the assessment of the current status of users and their entitlements from the existing IT environment (i.e., here are the permissions Fred has); grouping of users into common entitlement profiles for reporting (i.e., these people are all "Tellers") and granting of entitlements (i.e., please give new employee Fred the things a "Teller" has), and in some cases policy compliance (i.e., Fred is a "Teller" but his permissions deviate from those of the "Teller" role)
- Password management and synchronization
- Bi-directional change synchronization to ensure that changes made at the native platform level are automatically synchronized with the EUA tool

- Delegated user administration for enterprises wanting to distribute the user administration responsibility
- Workflow facility for automating the access request process, often through integration with Human Resources applications
- "Batch job" facility for automated and rapid establishment of entire groups of users (especially important for mergers, acquisitions and application conversions)
- Single sign-on, either through their own solution or through partnership with existing Single Sign-On (SSO) solutions
- Public Key Infrastructure (PKI) support for activities such as certificate management
- Customizable reporting, including a user view and platform view across the environment
- Audit log and reporting of sensitive activities — for the administration of the product as well as user administration.

Drawbacks to a successful Implementation EUA

The major drawbacks are the lack of documented access roles based on business functionality and weakly defined procedures for handling users' changing roles. Reorganization will always put a burden on security administration staff. At this point a tool that helps automation of changes helps a great deal. However, if the rules regarding access when a user moves from one job to another are ambiguous, not even an automated system tool can help. They only reveal the shortcomings in procedure that the organization must attend to.

In an attempt to solve some of these identity management initiatives, companies are implementing some of the solution alternatives that are discussed in this paper. The first step will be defining what is required of an identity management solution.

3.1 Identity Management Solution Requirements

Identity data will continue to exist in many places, and companies need to find ways to make different directory services and application repositories work together. Assuming that there will be many identity repositories, solutions must provide:

- Connect ability to the many forms of identity data.
- Management including failure management of identity flow between repositories. This is known as Brokering.
- Ways and means for maintaining integrity of data throughout the infrastructure used for identity management, including referential integrity.
- Change Event Processing.
- Aggregate the different data from the different repositories into a meta directory that will have a holistic perspective of identity data from the organization as a whole.

3.2 Connectivity

Connectivity needs directory services, databases, and applications to which identity management solutions can connect. The management solution will only be able to connect if it can:

- Obtain information about what has changed in the repository.
- Add new objects to the repository.
- Delete objects from the repository.
- Change an existing object's attributes to different values.

To be a comprehensive solution, technologies should be able to connect to data in:

- Standards-based directory services via LDAP Version 3.
- Popular existing e-mail applications and non-LDAP directory services.
- Enterprise Resource Planning (ERP) applications.
- Databases via access methods, such as structured query language (SQL).
- Applications in which the only interface to identity information is through application programming interfaces (API) and no directory interface is available.

3.3 Joining/Brokering

It is a process of managing the flow between repositories of identity information. This functionality must be able to:

- Sense changes to identity data and propagate updates to other repositories
- Aggregate data from different repositories into meta-directories that contain a holistic view of identity data from across the enterprise
- Track related objects as they change their positions in directory trees and other repositories due to periodic reorganization.

3.4 Change Event Processing

Whenever events administrators, users, or applications add, delete, or modify a piece of identity data in a repository an event happens. If this ability isn't functional as it ought to be that is to detect and process changes then data containing identity quickly gets disorganized. Solutions therefore must give features that enable the detection changes, perform necessary data format translations that make related updates to occur in all repositories that should reflect the change. An example, is if a person is added to the HR's database as a result of

new hiring, this change event must cause systems that the person will use to reflect the addition.

3.5 Data Aggregation Capabilities

Meta-directories that contain an aggregation of identity data from many other repositories can offer great value though identity information resides throughout most enterprises. For example, applications can access a wide range of information in one location with a single access means and security model instead of having to interact with all of the source repositories directly. Because has information stored in an indexed manner, performance is maximized and there is no need collect data from sources, which may reside across wide area network (WAN) connections, at run time. In order to do this, data aggregation capabilities must be able to:

- Collect and incorporate data from many sources including directories, databases, and applications.
- Categorize related data together, even though it may be stored in different ways in different places. For example, a user named John Smith might have data stored under names such as *Chris Angura*, *cangura* and *angurac* in different systems.
- Send the changes back to sources when users or applications make changes to the aggregated view. This means that meta-directories must be integrated with change-event processing infrastructures.

3.6 Related Object Tracking

Identity management solutions on first deployment by administrators initially must be able to provide ways for establishing relationships between related pieces of identity data stored in different repositories. For example, administrators must be able to tell the broker that *Chris Angura*, *Cangura* and *Angurac* are all the same person. It must be able to track relationships as identity data is reorganized periodically. For example, solutions must not lose track of users simply because they change positions in a directory tree structure due to a move from the Accounting department to the Sales group.

3.7 Integrity Management

It is the process making sure that identity data is not corrupted or out of synchronization between repositories as changes and brokering processes occur. Integrity management functionality must be able to:

- Maintain ownership of identity data relationships
- Take appropriate when failures occur
- Maintain referential integrity between identity data

3.8 Ownership

A very vital part of enterprise identity management is noticing that there are important ownership relationships to be kept between applications and data. For example, a person's mailbox name is owned by the e-mail system that hosts the mailbox. Within most organizations, the human resources (HR) system owns the data corresponding to whether or not a person is an active employee. Without an enterprise identity management infrastructure, these relationships are preserved by default because no other applications have the ability to access and update e-mail and HR data. With synchronization connectors and brokers deployed, however, the situation changes.

For example in the case where mailbox information is being synchronized to the human resources (HR) directory by a connector. If the connector is not configured correctly, a user could change the mailbox attribute in the HR system and the connector would overwrite the mailbox value in the e-mail directory – causing tremendous confusion.

Solving the problem is not as simple as just preventing changes from flowing backwards to the e-mail directory. The HR system may be the owner of information such as the name of a person's manager—that needs to be flowed back to the directory in the e-mail system. Other attributes, such as a person's office number, may have no clearly defined ownership and should be data that anyone can update.

As a solution requirement, connectors and brokers must enable administrators to define and enforce ownership relationships at the attribute level. In this way, if a change observes ownership rules, connectors or brokers allow it to pass through. If an administrator mistakenly changes a mailbox attribute in the HR directory, the identity management solution would simply set the attribute back to the value contained in the e-mail directory.

3.9 Failure Management

The capability to send a change to multiple repositories is a key requirement for brokering technologies. However, any time a broker makes multiple updates, the chances for one or more of the updates to fail exist hence making data inconsistent in different repositories. For example, if a person is added to the HR system and the broker is unable to contact the e-mail system to add a mailbox, identity data will be left in a state of confusion. Typically, this means that an administrator will have to investigate the situation and make corrections by hand. To combat these kind of failures mechanisms that ensure that all updates occur successfully or are rolled back as a unit in database systems are instituted.

3.10 Referential Integrity

Maintaining *referential integrity* between repositories is a big challenge. This refers to the need to maintain relationships between the values of related pieces of data in different locations. For example, identity management solutions must be able to ensure that a person's title listed in the human resources (HR) system is consistent with their spending limit in the procurement system. Databases help solve this challenge by providing stored procedure and trigger features that enable administrators to execute a business rule each time a data value changes. As directory services do not provide similar features today, identity management

solutions must provide the capability to execute business rules within brokering engines and reject changes that do not meet referential integrity requirements.

3.11 Challenges of Identity Management

Due to the different types of identity data and the number of places where such data reside there are several management challenges that have been posed today:

- Different ways of representing Identity data, (data type, data format, data value), for the same user. i.e., Chris Angura, cangura, cangura04, achris. (all the same person)
- Not all identity data is kept in directories or exposed through a directory service interface such as LDAP. For example, many systems only expose identity information through specialized application programming interfaces (APIs).
- No common directory schema agreed upon by vendors
- Applications can't easily be modified to use directories. Most applications do not support standards or integrate with a directory
- Drift out of synchronization over time if left unchecked if identity information is duplicated in multiple places, and versions frequently
- There is no single place where administrators and applications can access or manage an aggregated view of an enterprise's identity information.
- The number of places where companies must manage identity data increases with each additional application and platform.
- Simple synchronization tools that some vendors supply are inadequate and do not provide sufficient provisioning capability. Most only work between two systems and do not cover all identity attributes.
- Data ownership and political issues
- Dirty data or data is incorrect.
- Immature standards.

These challenges mentioned above have made it very hard for companies to implement comprehensive and integrated identity management solutions. This has increased costs and complexity.

3.12 Value of an MDS Identity Management Solution

“The aggregate directory, Meta Directory, produced by the Meta Directory Service (MDS), provides the rationalized and unified view of identity objects that consist of attributes from a variety of directories and identity repositories. This integration will lower administrative costs, eliminate duplication, reduce discrepancies, and make the identity information widely available.”

3.13 Lower Administration Costs

A Meta Directory can be used to cut down the cost of user and account management by automating account creation, deletion and management processes. The integration of data from authoritative sources, such as Human Resource Systems or Customer Databases, enables event driven account management so that user accounts can be created and maintained automatically. An enterprise will be able to hire a new worker or give an invitation to a new business partner to share their network resources with knowledge that everything that person needs to be productive will simply and automatically be delivered to the right person at the right time.

3.14 Provisioning

Customers facing an increasingly hard task of adding, moving, and changing availability of the resources to support the business activities of employees, contractors, business partners, and vendors. Customers must provision voice and data network resources, security systems, remote access systems, operating systems, applications, Web-based information services – in addition to services that are outsourced to traditional outsourcers or Application Service Providers (ASPs). This is a very key as one of the requirement in business and very important benefit a Meta Directory solution will provide.

3.15 Reduced Security Risks

In case there are changes in the status of an employee a Meta Directory helps to reduce security risk by providing a way for the organization to simultaneously disable access to all systems used by the person based on employment or account status change. Also, employees can more easily access systems by using a single userid and password as created and propagated by the Meta Directory through automated account creation and management

3.16 Accelerate Standard Business Processes

An access to directory information that is unified ensures that e-business customers, partners, and suppliers have access to consistent identity information. This is achieved when a Meta Directory makes identity data available in a unified view by integrating data from a variety of sources into a Directory Server. It also enables a unified view of a person's accounts and profile information by joining account information from multiple sources, such as databases, HR systems, network operating systems, messaging systems, telephone switches as well as other directories, into a single profile. This unified view, can then be used by other applications that are directory aware.

3.17 Integrates Messaging System Address Books

Integration of multiple messaging system directories can also be achieved with a Meta Directory so that all email addresses throughout the organization are easily visible to any user on any of the messaging systems, migration of account from messaging system to system by providing accurate email addressing before and after account migrations. In addition to that, the joining of accounts from a HR system with the messaging systems will enable the creation of a business card email address that will not change even through account migrations.

3.18 Approaches to Identity Management Solution

Following up what was mentioned earlier, there are many ways to address identity management issues and here is where we will discuss some of them. Businesses need to examine their environments to assess what solutions are best suited for them and what the cost/benefit are. A MDS directory solution has numerous advantages and gives an all round solution. There may be situations where some of these solutions may be enough for some of the identity repositories and may work alongside a more complete Meta Directory solution.

3.19 Home Grown Replicate from non-directory source to directory

The commonly attempted form of a Meta Directory is replication, which involves copying attributes. Here data is copied from its source location into the Meta Directory using in-house custom extracts and populates processes. This is a homegrown solution form of MDS. For improved performance and manageability data can be extracted from a very limited source and restructured. This has an advantage of the ability to use existing directory technologies.

The disadvantages, however including limited scope of a stovepipe solution. The system just copies data, without any application of any additional intelligence to take into account other systems that are trying to perform the same function. Here, there isn't any intelligent

join mechanism to put all records together. This solution does not meet the requirements of an Identity Management solution we have stated earlier.

3.20 Direct Access to Each Repository

One way to manage multiple sources of identity data is to write applications that interact directly with each of the repositories that hold identity data relevant to a particular management operation. For example, administrators could implement a hire & fire solution by writing a script that, when an employee joins the company, adds appropriate identity information to the HR system, network operating system directories, and each application that the employee will need to use.

This approach requires developers to learn how to access each repository. Applications and scripts can contain business rules for updating data and attempting to maintain consistency across repositories. No synchronization of changes is needed as scripts and applications update all repositories directly. This approach does not create an Enterprise Directory where applications, users, and administrators can go for a holistic view of identity data. Business rules are not easily maintainable and updateable. Also, when other applications make changes in individual repositories, there is no mechanism to propagate the changes to other directories, databases, and applications affected by the change and ensure they are following corporate policies set forth in the business rules.

3.21 Brokering of Information

Another technology where the directory server does not keep copies of attributes locally called the information brokering or chaining. The information is retrieved using indices or pointers from their respective proprietary sources and the values are returned to the requester using standard directory protocols. This has advantages of this approach are elimination of synchronization delays, resulting in the most up-to-date information and minimum storage requirements since there is no duplication of data.

However, there are disadvantages which include poor search performance because of the difficulty of building indices for linked or dynamic attributes. Also there is unreliability of the server since the Meta Directory server relies on the network and the connected directory for availability before generating requested data.

3.22 Directory Consolidation

If identity data is consolidated around strategic directory services there will be a big return on investment. This is because there will be a decrease in the number of directories where administrators must store and maintain identity data. There is a growing trend among software vendors to consolidate identity repositories amongst their own product lines. However, most vendor applications are still written against their own APIs to go against their own directories. Directory objects and attributes have not been standardized throughout the industries and company politics make a single directory very difficult.

3.23 Challenges of single Identity Management repository

Many companies have tried to create a one single directory that will hold all the enterprise identity information. However, most of these efforts have failed due many factors::

- Many applications cannot be modified easily to use directories.
- There are good reasons, such as various replication and security requirements, why some applications need to keep identity in their own formats.
- Political boundaries make it hard for complete consolidation, regardless of what is technically possible.

3.24 Most comprehensive enterprise identity management solution

Hub and spoke architectures represent the best approach to building comprehensive enterprise identity management solutions. Critical Path, iPlanet, Microsoft, Novell and Siemens all provide a hub and spoke architecture. This is the type of solutions used in MDS.

4.0 Available Vendor MDS Solutions

As of the writing of this paper, the following is a subset of current vendors' MDS solutions.

4.1 Novell's Solution

DirXML™

DirXML™ is Novell's innovative data-sharing and synchronization solution. DirXML™ works with an existing network infrastructure, automatically distributing new and updated information from any specified directory across all of the directories, applications, and databases. When DirXML™ is used; the information in all of the data repositories can be simply updated and instantly synchronized.

DirXML™ is also a cross-platform product that uses together the power of NDS® eDirectory™ with the flexibility of Extensible Markup Language (XML). Designed with e-business in mind, DirXML can even work beyond your firewall, updating information in databases shared with customers, partners, and suppliers.

Benefits

- Update every directory on your network with a single change to one directory, application, or database
- Enjoy dramatic savings in the amount of time and money spent updating information
- Resolve data ownership concerns with publisher and subscriber channels

- Ensure that e-business customers, partners, and suppliers have access to consistent directory information
- Eliminate expensive, outdated hard-copy directories
- Accelerate standard business processes
- Leverage NDS eDirectory to enjoy platform independence
- Use XML to communicate with every directory on your network.
- Use the included, application-specific drivers for some of the industry's most popular directory applications
- Develop customized C++ and Java drivers with the included Software Development Kit (SDK)

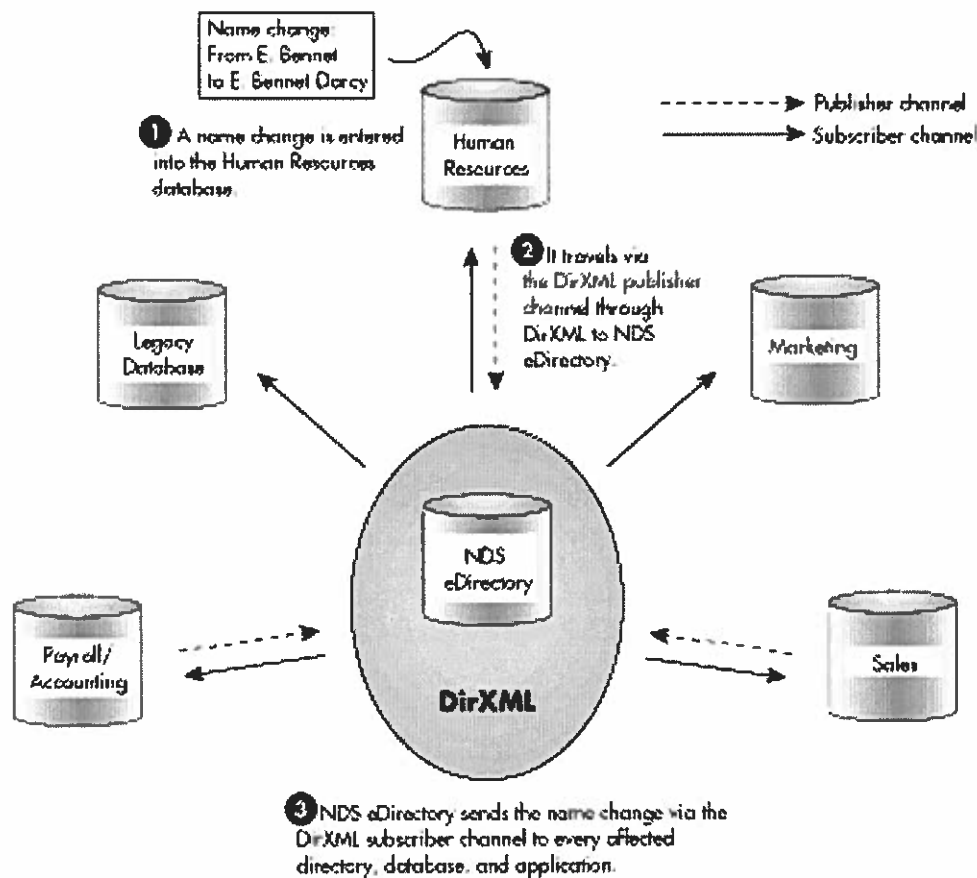


Figure 1. DirXML publisher and subscriber channels update information in all the databases on a network

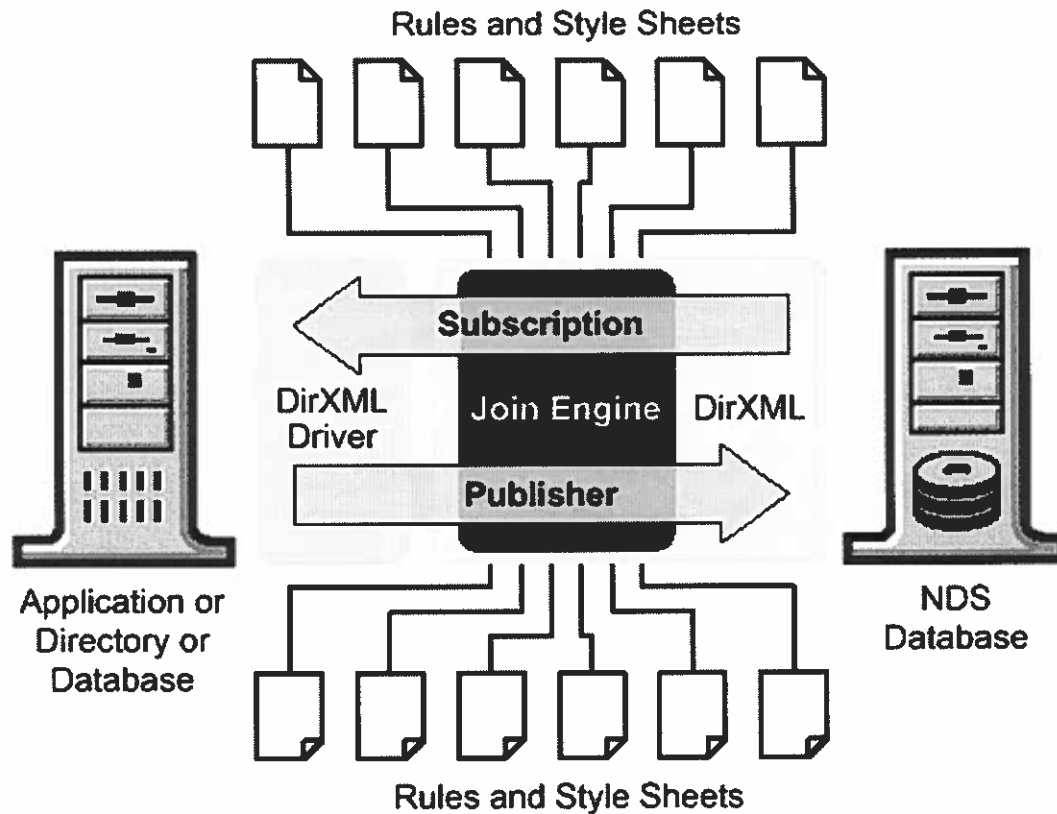


Figure 2. Shows the basic DirXML components and their relationships.

4.2 Microsoft Solution

Microsoft Meta Directory Services (MMS) (Beta)

According to Microsoft hub and spoke architectures represent the best way of approach in building a comprehensive enterprise identity management solution.

Zoomit, the industry's leader in Meta Directory technologies was acquired by Microsoft Corporation, Zoomit's flagship VIA product, renamed Microsoft Meta directory Services (MMS), implements a hub and spoke architecture.

MMS enables companies to utilize Active Directory to manage identity information stored in directory services. That are not homogeneous. Microsoft also is improving MMS to become the industry's first comprehensive enterprise identity management platform by adding support for many popular applications and network services that store identity information in places other than directories.

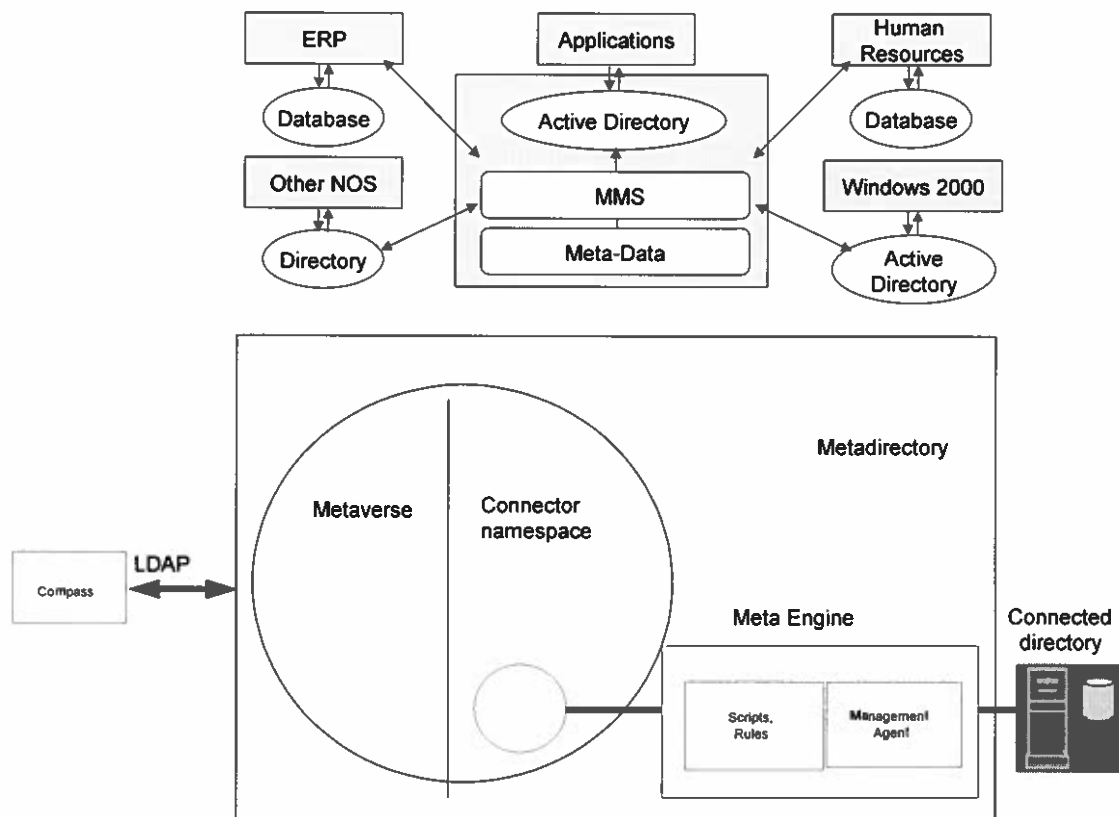


Figure 3. MMS Architecture

Note: Microsoft's LDAP is implemented on top of Active Directory (AD) which gives Microsoft the freedom to decide which protocols and services can get access to data stored in the directory. Fully access of Active Directory, got through ADSI -- Microsoft's proprietary Active Directory Services Interfaces. Conversely, when using Active Directory APIs, no LDAP operations are used, which means developers are locked into using ADSI.

4.3 Critical Path's Solution

InJoin™ Meta Directory (formerly ISOCOR MetaConnect)

MetaConnect is based also on the integration of multiple data sources into an LDAP directory managed by the MetaConnect meta-directory. An exchange of information between MetaConnect and the data stores for a variety of products is done through a set of connectors. These connectors form a ConnectorView for each data source. The Join Engine links selected entries from the individual data sources into ConnectorViews. These ConnectorViews are then used to create a single consolidated view of the information in a MetaView. The MetaView is the central repository of all of the information and is maintained in an LDAP directory. MetaConnect also provides management and

configuration GUIs to allow organizations quickly and easily to determine how information should flow through the data systems in their environment.

because MetaConnect is directory server independent it doesn't need additional directories.

The JoinEngine can build and manage the Meta Directory in any LDAPv3 compliant directory server with selected LDAP extensions to the server or to the JoinEngine. By doing this, it enables the use of an existing LDAP directory server to provide the Meta Directory store, rather than calling for the installation of another directory server.

Version 2 also includes full support for Microsoft's Active Directory. Customers can use the Microsoft Active Directory as the repository for all MetaConnect information. In this configuration, MetaConnect does not require any other directory services. MetaConnect version 2 can also fully manage Windows 2000 users and groups through the Active Directory connection.

System Requirements

InJoin™ Meta Directory consists of a range of product components, each with different system requirements.

InJoin™ Meta Directory JoinEngine (MetaConnect V2) is available on Windows NT®4.0, Windows 2000 and Sun Solaris™2.6.

Platform support for the connectors varies.

The Connectors

This is used to manage the interfaces between MetaConnect and the individual data sources. Each ConnectorView contains information about a single data source and provides the necessary protocol conversion for the exchange of information. Supported environments include:

• Lotus Notes	• Oracle
• Microsoft Exchange	• Sybase
• cc:Mail	• SQL Server
• Fischer TAO	• DB2
• N-PLEX	• Windows NT
• NetWare	• DB2

Critical Path also provides connectivity toolkits for accessing SAP and PeopleSoft data. Additional connectors are being developed constantly: check with Critical Path for the latest information.

Connectors can also be used to exchange information between the tools in a more direct fashion, and this capability becomes more important as we move towards the management and synchronization of directory-based information.

In addition to connectors for common messaging systems, there are similar connectors available for network operating systems and databases. The Universal Database Connector provides a direct view of the contents of a database, while the Universal Connector makes use of replication to provide information to the ConnectorView. In this latter case, Perl script can be used to perform data management functions during the replication process. This combination of direct and indirect access to the underlying directory enables the features of relational databases to be used, and also gives flexibility in how the Meta Directory contents are handled.

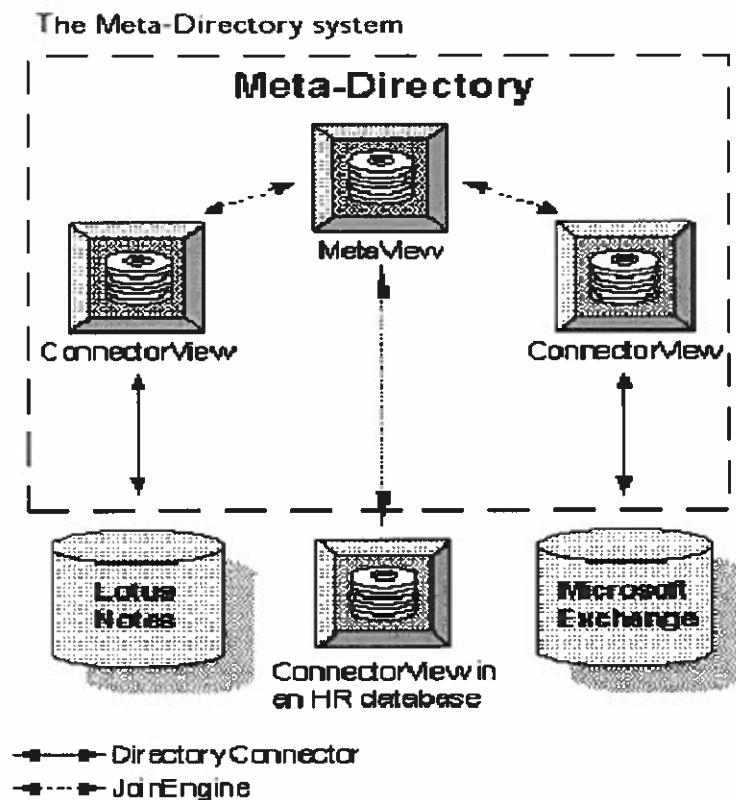


Figure 4. The Meta-Directory system

4.4 iPlanet's Solution

iPlanet Meta Directory v 1.x (formerly ISOCOR MetaConnect)

iPlanet Meta Directory v1.x is also ISOCOR MetaConnect. iPlanet purchased ISOCOR MetaConnect product. This was done before CP acquired ISOCOR. The functionality should be the same, the documentation is. It was like déjà vu.

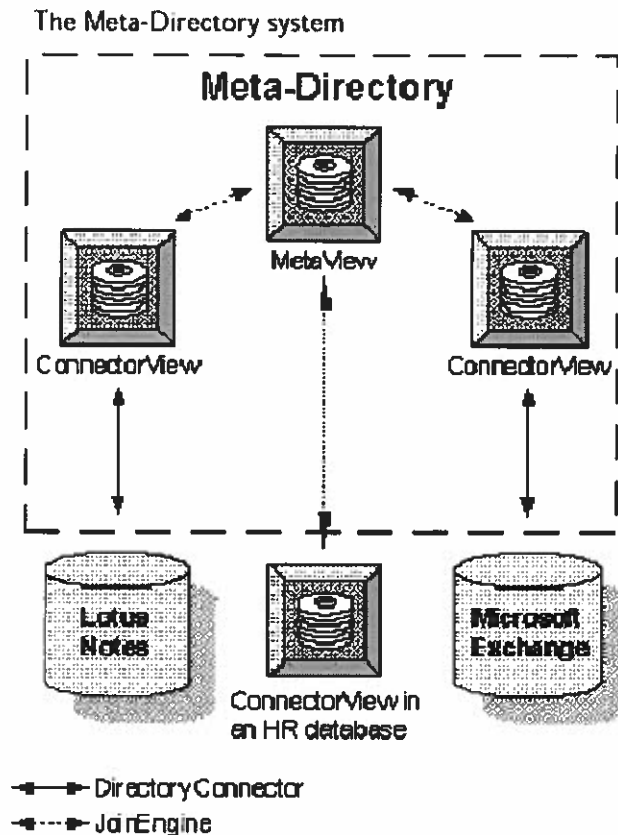


Figure 5. The Meta-Directory system

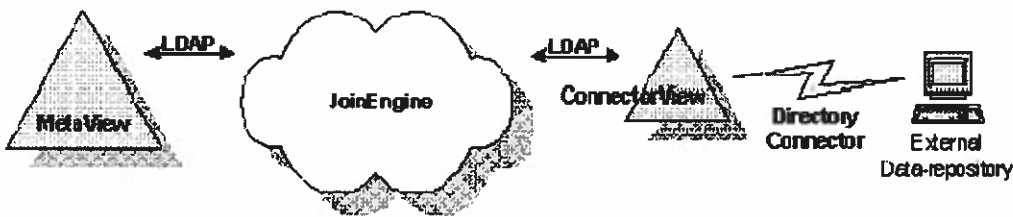


Figure 6. Data integration into Meta-Directory

5.0 Deployment Process

In implementing an identity management solution the first step is identifying business objectives, policies and strategies both current and future. Next, a complete evaluation of the identity repositories, current identity management implementations (i.e., synchronizing messaging and desktop Ids), political boundaries and current vendor capital investments and loyalty. This is help to ascertain the type, value and scope of the solution. Many companies have looked at parts of their IT infrastructure, i.e. Messaging and Network Operating System (NOS), but have not taken a holistic approach, one which includes applications and HR. Business case including Total Cost of Ownership (TCO)/ROI and value analysis should follow.

This should be a holistic approach, but that does not mean the service can not be implemented over time. Because of the size of this task, it is suggested that the following be considered.

- Start small.
- Grow over time.
- Deploy in parallel with existing systems.
- Provide an Enterprise Application Integration (EAI) Solution.

Translating data and commands from the format of one application into the format of another.

Most of the practices that are involved in implementing a Directory Server will apply since the backbone of all MDS is the Directory Server.

Key issues that are very important to take of:

- Study Client's business environment.
- Know corporate structure.
- Survey the enterprise and identify where the data comes from (such as NT or Netware directories, PBX systems, Human Resources databases, emailsystems, and so forth).
- Locate all the organizations that manage the enterprise's information.

- For each piece of data, determine who owns the data; that is, who is responsible for ensuring that the data is up-to-date.
- For each piece of data, determine the location where it will be mastered.
- Identify the tools and processes that the enterprise uses to maintain this information.
- The decision about what types of data are maintained in the directory, and when you will start maintaining it there, will be driven by several factors:
 - The data required by the various legacy applications (such as existing email applications) as well as the user population.
 - The ability of the legacy applications to communicate with an LDAP directory service.
- For each piece of data, determine the name of the attribute that you will use to represent the data in the directory and the object class (the type of entry) that the data will be stored on.
- Determine how centralizing each piece of data will impact the managing organizations.
- Determine what future directory-enabled applications that will be deployed and what their data needs are.
- If a unique identifier is not ubiquitous in the organization, the meta directory service should begin the process of imposing one.
- Obtaining a corporation Object Identifier (OID) Tree from IANA that can be used to uniquely identify attributes and object classes.
- *Best practices* in directory information tree layout include having as flat a directory tree as possible. Note that this requirement is one of the best reasons why the NOS DIT and the enterprise directory DIT, and therefore directory, need to be different.

5.1 Information Gathering

Assessment of Current Business Policies.

The main bottlenecks to implementing a successful Meta Directory solution are the lack of precisely documented access roles based on business functionality and weakly or not defined procedures for handling changing roles of users. In order to overcome this potential problems it is necessary to work closely with a corporations Security group to produce a Security Engineering Plan (SEP) to ensure that the proper procedures are well defined.

Assessment of Corporate Identity Environment

Identity repositories take on many forms. Typical identity repositories are RDMS, NOS and application specific repositories, (i.e., email).

All systems, which require access control, require an identity repository. Most repositories are meant solely for use by the application storing the information. Yet others are used for multiple purposes, such as Directory Servers. Most identity data is redundant. The semantics are the same though the syntax and data definitions vary between systems.

The AttributeWorksheet.xls is a tool that maps identity attributes. The matrix should list each repository, its use (i.e., email), and the name and type of each peace of identity

information, (attribute), stored within. This tool can assist you in normalizing, associating your identity data.

6.0 Assessments

Assessment of Partial Implemented Identity Solutions

As mentioned earlier in this paper there are a number of identity management solutions. One form of identity management that some companies have used is synchronization connectors. Some vendors supply synchronization connectors with their products. These synchronization connectors are applications that know how to recognize changes in one repository and propagate them to another repository. Connectors typically maintain a one-to-one relationship and companies need to have a specific connector for each pair of repositories that will be managed.

It should be noted what systems are currently in place to help address Identity management and who a Meta Directory will interact/replace such a system.

Assessment of Political Boundaries

Because of the number of organizations that can be affected by the directory, it may be helpful to create a directory deployment team that includes representatives from each affected organization. Including representatives from each of these organizations can help to more rapidly perform the survey. More importantly, it always helps to listen to the users. Directly involving all the affected organizations can go a long way to building acceptance for the migration from local data stores to a centralized directory service. This centralization will require adding personnel to some organizations while reducing head count in others. The net reduction in administrative task will free employees to move on to more important tasks. They may not realize this at first and be somewhat threatened. It is important to present the opportunities you are opening up for your former administrators.

Assessment of Vendor Loyalty and Investment

Companies may have large investments in iPlanet, Novell, Microsoft or IBM. It would not make business sense to bring in a new Directory vendor into the mix if a current vendor solution will do. Some vendors are Directory independent, which allows a company to leverage existing resources.

Analysis of Information

The foundation of all decisions should be based on business and security principles. The analysis should produce a Policies and Security Engineering Plan. Once you have established your business rules and policies, and located all the data important to the enterprise, it must be determined if the data really can or should be stored in the Meta Directory, who is the Master, Peer, or Slave and if any translation is needed. The various vendor products all supply analysis tools to help in this process. Knowledge comes with experience and sound-consulting practices, such as those practiced at CSC.

The Attribute matrix, mentioned earlier, is a tool I have used to help me show the relationships of attribute data and was derived from various vendor tools that where not quite in themselves what was needed.

The CSC/TMG System Architect groups can assist in the analysis phase.

Implementation Process

Implementation is vendor specific and based on the technologies deployed by each vendor.

The degree of functionality as discussed in section "Requirements of a Meta Directory solution" is also vendor specific.

It is best to consult the vendor documentation for detailed information.

7.0 Architecture

Each vendor's MDS product implements its architecture based on the functionality of its components. MDS vendors provide high-level conceptual diagrams of there solutions. It is hard to accurately document the architecture of a vendor's MDS solution unless you have used and understand the solution. Of the solutions the paper addresses , Novell's DirXML was chosen because of it's event driven and XML, XSLT data translation capabilities. Therefore I present the DirXML architecture.

Novell's DirXML

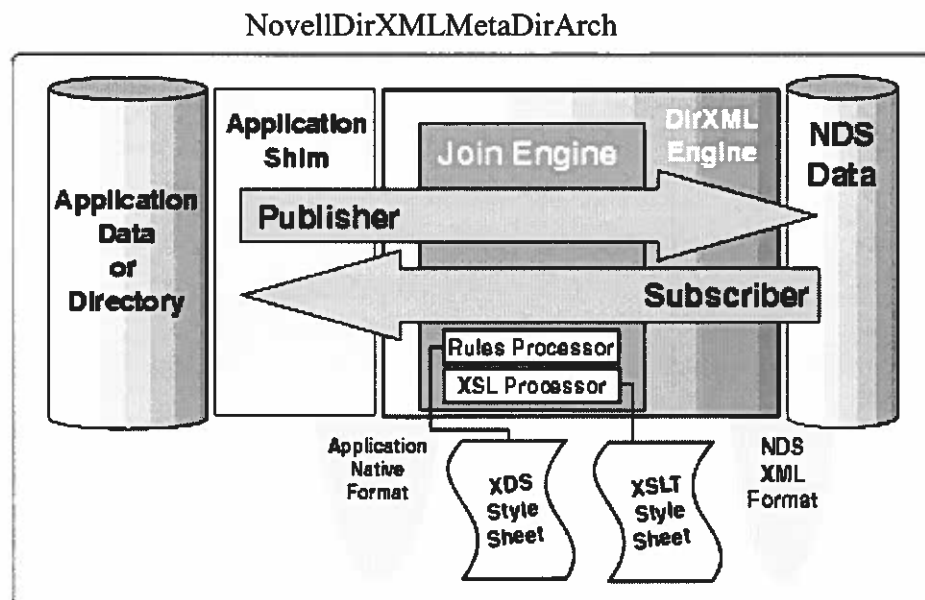


Figure 7. DirXML Architecture

The Subscriber and Publisher channels communicate with NDS eDirectory through an *Application Shim*. An Application Shim is an application-specific piece of code, which knows how to communicate with the target application. The shim passes data to the application once the data has been processed by DirXML and receives data from the application that is translated before it goes to NDS eDirectory.

The *DirXML engine* is a collection of interfaces and data manipulation technologies that allow even disparate data systems to connect and share data. The DirXML engine provides an interface to NDS that exposes NDS data and NDS events using an XML format. The DirXML engine employs a rules processor and a data transformation engine to manipulate the data as it flows between two systems (see Figure 3). It also provides an interface for application-specific programs to attach through the DirXML engine to capture the data streams that flow from NDS.

NDS eDirectory is a directory service that manages data between multiple servers. When changes occur to the data, all other servers which hold copies of that data are informed of the change and make corresponding updates.

When DirXML is loaded on top of NDS eDirectory, these events are passed between NDS and the DirXML engine. NDS manages the drivers that are loaded and configured to consume these events. NDS also provides some basic management services related to the events themselves so that events are always stored until they've been successfully consumed.

The DirXML engine converts the NDS events into XML documents (in DOM format). Rules are applied to the XML document and data transformations are performed to convert the data into the target application's native data format. Once the XML document has been completely processed, the event data is handed to an Application Shim which delivers the formatted data to the target application. We'll explore each of these steps in detail:

DirXML is represented by objects defined in NDS. The base NDS schema has been extended to accommodate this information. The new object types are:

- **Driver Set.** An object that defines a collection of drivers.
- **Driver.** An object that defines all the components of a driver including rules, style sheets, and the Application Shim.
- **Rule.** An XML document that defines a rule that will be applied to the NDS event stream as it flows through the DirXML engine. Examples of these rules are Event Rule, Schema Mapping Rule, Matching Rule, Create Rule, and Placement Rule. These will be discussed in more detail below.
- **Style Sheets.** An XSLT document that defines a transformation of the NDS XML code into some other data format.
- **Application Shim.** Although not specifically represented as an object in NDS, the Application Shim is worth noting as it completes the driver definition. The Application Shim is an executable piece of code that the DirXML engine uses to interface with the target application. This is actually stored as an attribute of the driver objects stored in NDS.

In general, the DirXML engine controls the flow of data between NDS and the target application. NDS provides the engine for storing the configuration and the event system that drives the entire process. The DirXML technology includes all the components necessary to move data between NDS and any given application.

Conclusion

With the complexity and cost involved in an enterprise identity management and the need for a corporate directory to implement B2B and B2C solutions, there should plans by companies for deploying a Meta Directory Service (MDS) solution or a combination of identity management solutions that addresses the specific needs of the company.

Due to Directories proliferation companies have become very dependent on each other, B2B, B2C, and the internet thus making a corporate directory a necessity. Thus, the tool to build that corporate directory is also a necessity.

MDS solutions offer considerable power and flexibility, and therefore the chance to solve big identity management challenges. Over time, current and maturing MDS solutions will provide the type of functionality outlined earlier in this paper and they will be easier to implement as more MDS providers roll out advanced capabilities such as templates, simplified rules languages (like XML), standardized data transformation mechanisms such as XSLT, and pre-built connectors to popular applications.

Implementers also must learn how to work with clients on defining the business rules/policies that will govern how identity management will be enforced. MDS do a fine job of automating well-defined procedures, but they do a poor job of clarifying poorly defined procedures.

MDS Solutions implement business policies. More importantly, they enforce business rules and provide central administration to ensure an enterprise focus.

The number of relationships MDS manages between applications with a dissimilar data store and the number of applications that can leverage the corporate directory produced measures the value of a MDS solution. Ultimately, the greatest return on investment comes from consolidating identity data around strategic directory services. This is because the number of directories where administrators must store and maintain identity data actually decreases. Synchronization is unnecessary. Software vendors have been consolidating identity repositories amongst their own product lines.

On the bright side, MDSs provide a mechanism for translating attributes and data to cover the case of synchronizing all the non-standard schemas.

Although MDS do create an aggregate data repository, Meta Directories, of all the identity repositories they synchronize to, they do not create a single directory that all the applications can use, as mentioned above. However, a business can begin the process of creating a directory that they can start to consolidate their applications around using MDS as a tool to accomplish this. The corporate directory should not be the MDS vendor's directory where the aggregate data is stored.

Directory consolidation is the goal and the corporate directory you select should provide as open architecture as possible, to allow various applications to integrate, and be supported on as many platforms as possible.

A Meta Directory service is a service that will enforce business policies, synchronize data repositories, and consolidate data into a single repository (Meta Directory). Although some MDS solution vendors want to make the directory they use to store the aggregate identity data, the "Corporate Directory", I would like to stress that the Corporate Directory be separate from the Meta Directory. The Meta Directory service will enable the creation of a Corporate Directory and provide a focal point and authoritative source for other directories required in an IT infrastructure but its repository should not be the Corporate Directory.

Since each leading vender has a directory that is integrated into its product line, it makes sense to consider leveraging the platform that a majority of your applications use for the Corporate Directory and this may or may not be the MDS vendors' directory platform.

References/Works Cited

LDAP Organizations

- Active Internet Engineering Task Force (IETF) Working Groups

Look for LDAP and XML activities

- Innosoft's LDAP World
- The Open Group Directory Interoperability Forum
- The Directory Interoperability Forum
- Active Digital Profile (adpr)
- CSC Sources - Communities of Interest - Directory Services

XML Organizations

- <http://www.w3.org/http://www.w3.org/>
- XML.ORG

NetworkWorldFusion

- **Sizing up LDAP servers**

Meta Directory Solutions

Novell

Novell DirXML

Microsoft

Microsoft Metadirectory Services (MMS)

Microsoft Metadirectory Services (MMS) Concepts and Architecture

Windows 2000 Interoperability

InJoin™ Meta-Directory

http://www.cp.net/products/injoin_metadir_overview.html

Siemens

Siemens Information and Communication Networks

Siemens Meta Directory

Glossary

- | | |
|---------|--|
| 1. LDAP | Light Weight Directory Access Protocol |
| 2. OS | Operating System |
| 3. DBMS | Database Management systems |
| 4. ERP | Enterprise Resource Planning |
| 5. EAM | Extranet Accesses Management |
| 6. EAU | Enterprise User Administration |
| 7. SSO | Single sign On |
| 8. PKI | Public Key Infrastructure |
| 9. MDS | Meta Directory System |
| 10. NOS | Network Operating System |
| 11. DIT | Directory Information Tree |
| 12. DEN | Directory Enabled Network |
| 13. EAI | Enterprise Application Integration |
| 14. PDC | Primary Domain Controller |
| 15. CD | Corporate Directory |
| 16. WAN | Wide Area Network |
| 17. XML | Extensible Markup Language |